

packages/graph-core/src/analyzer/language-lesson.ts

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Overview

`packages/graph-core/src/analyzer/language-lesson.ts` is a new module that adds a lightweight LLM-driven lesson generator for graph nodes. It introduces a `LanguageLessonResult` interface, utilities for detecting language concepts, prompt construction, and helpers for extracting and parsing JSON from LLM responses.

Key changes

- **Imports (lines 1-2):** `GraphNode`, `GraphEdge`, and `LanguageConfig` types are pulled into the analyzer module.
- **`LanguageLessonResult` (lines 4-7):** Declares `languageNotes: string` and `concepts: Array<{ name: string; explanation: string }>` for the lesson output.
- **Base concept patterns (lines 13-42):** A static map of common language concepts that is merged with language-specific concepts from `LanguageConfig`.
- **`buildConceptPatterns` (lines 48-63):** Merges base patterns with optional `langConfig.concepts`, ensuring each concept has a keyword list.
- **`detectLanguageConcepts` (lines 69-92):** Scans a node's tags, summary, and `languageNotes` for keywords, returning matched concept names.
- **`getLanguageDisplayName` (lines 99-107):** Returns a user-friendly language name, falling back to capitalization.
- **`buildLanguageLessonPrompt` (lines 112-155):** Builds a prompt that asks an LLM to produce a lesson, including node metadata, relationships, and detected concepts.
- **`extractJson` (lines 160-172):** Strips markdown fences or extracts the first JSON object from an LLM response.
- **`parseLanguageLessonResponse` (lines 178-210):** Safely parses the LLM output into a `LanguageLessonResult`, providing defaults on failure.

Impact

- **Correctness:** Adds deterministic concept detection and robust JSON extraction, reducing parsing errors.
- **Maintainability:** Centralizes language-specific logic; new concepts can be added via `LanguageConfig`.
- **Performance:** Linear scans over node text; negligible overhead for typical graph sizes.
- **Compatibility:** No API changes to existing modules; the file is purely additive.

Risks & follow-ups

- **`LanguageConfig` availability:** Verify that the type is exported from `../languages/types.js` and that consumers provide it when needed.

- **Prompt length:** Unknown from the available diff/scan evidence whether the generated prompt stays within LLM token limits for large graphs.
- **JSON extraction edge cases:** Unknown; add unit tests for responses with nested code blocks or malformed JSON.
- **Integration testing:** Unknown; confirm that downstream consumers correctly handle the new `LanguageLessonResult` shape.